Task Report

Parity Cube Assignment

Vikas Rajpurohit

02nd April, 2025

task one1 . : Write a Bash script (current_time.sh) that: Prints current date and time every 30 seconds. Appends the output to a log file (e.g., /var/log/date.log). Run current_time.sh continuously in the background. Ensure it restarts on failure. Starts automatically on server reboot.

Completed :

Here is shellscript where i have written logfile location as mention in task after that create file and give permission using while loop print date and time in log file sleep 30 after 30 sec it will print.

```
vikas@admin:~/parity_Assignment/shellscript_current_time$ cat current_time.sh
#!/bin/bash

LOGFILE="/var/log/date.log"

# Create the logfile if it doesn't exist and ensure write permissions
sudo touch "$LOGFILE"
sudo chmod 666 "$LOGFILE"

while true; do
    echo "Current Date and Time: $(date)" >> "$LOGFILE"
    sleep 30
done
```

I have move copy this shellscript to propar location in linux

```
vikas@admin:~/parity_Assignment/shellscript_current_time$ ls /usr/local/bin/
current_time.sh  irb  rackup  rails
```

To this location and server this shellscript by creating a service file for background running and if server get reboot then also it will auto restart or startupmode.

```
vikas@admin:~/parity_Assignment/shellscript_current_time$ cat /etc/systemd/system/current_time.service
[Unit]
Description=Log current date and time every 30 seconds
After=network.target

[Service]
ExecStart=/usr/local/bin/current_time.sh
Restart=always
RestartSec=5
User=root

[Install]
WantedBy=multi-user.target
```

Here is server file after that

sudo systemctl daemon-reload

sudo systemctl enable current_time.service

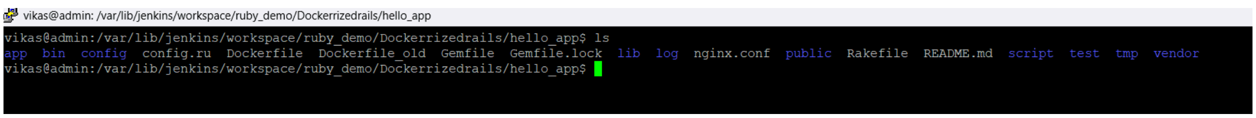
sudo systemctl start current_time.service

task two

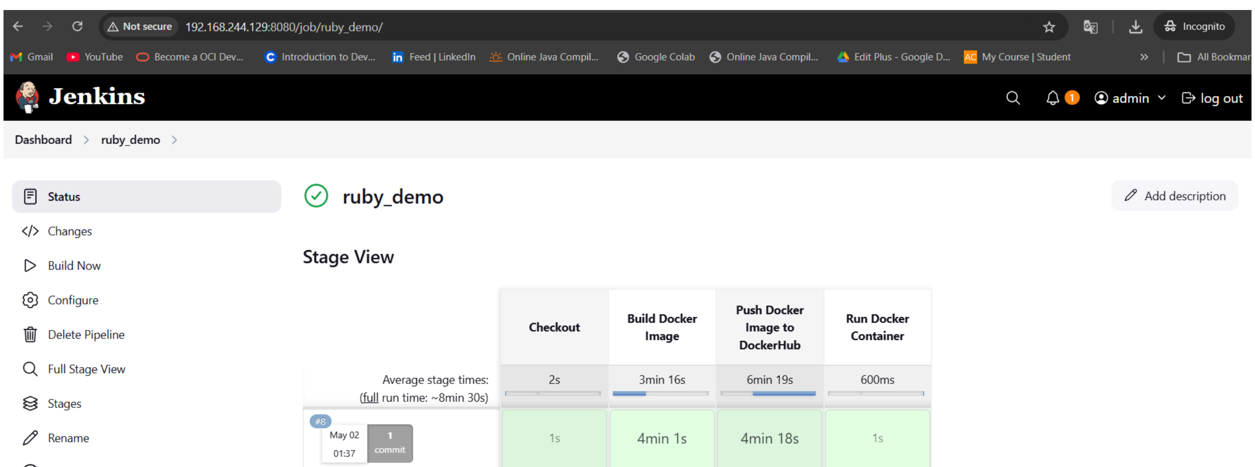
1. **Deploy a basic Ruby on Rails application that serves a "Hello World" message using Unicorn as the application server and Nginx as the reverse proxy, on an Ubuntu server.**
2. **dockerize this application , create docker image and docker container**
3. **create cicd pipeline using jenkins**

Completed:

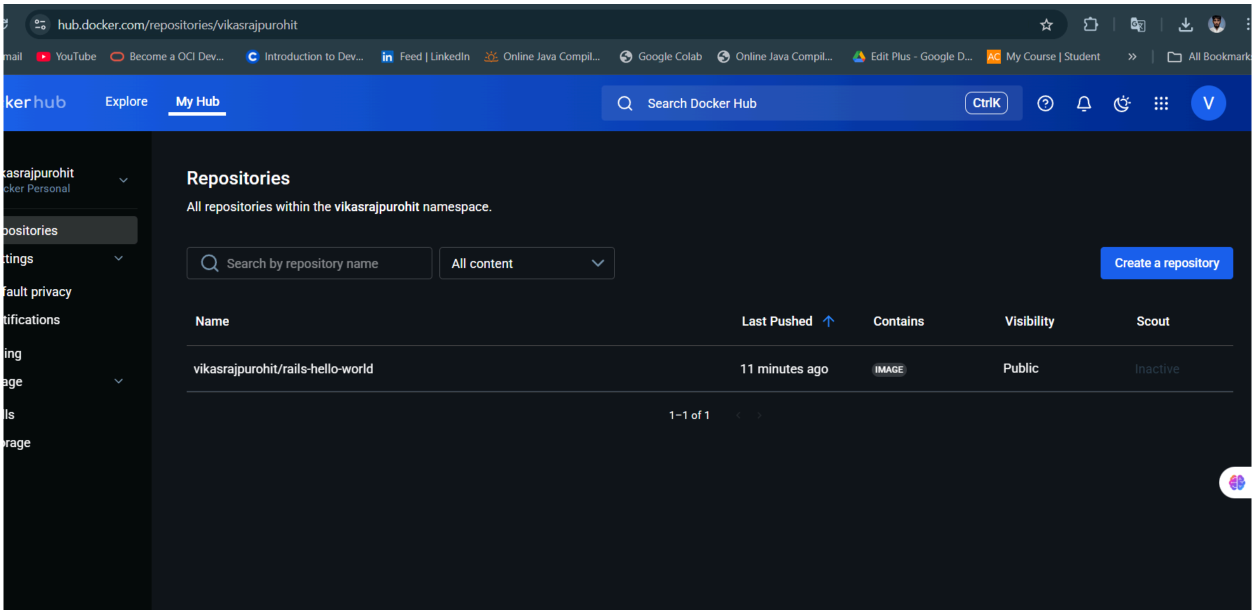
First did setup of ruby on ubuntu for project installation and running.
After that install project and install bundle also.



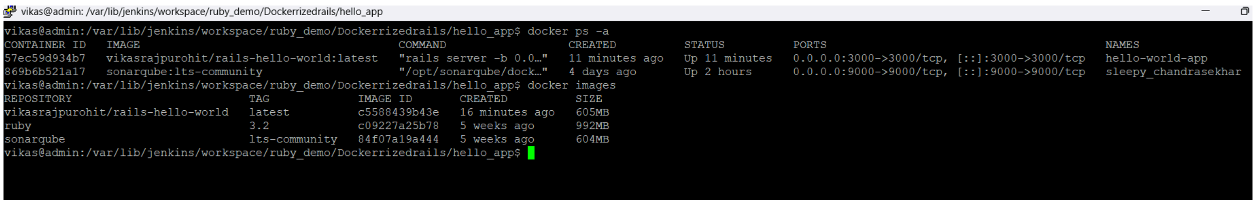
Here is project folder snapshot where also add unicorn in file in config folder and dependecies in Gemfile.
Also setup jenkins CICD



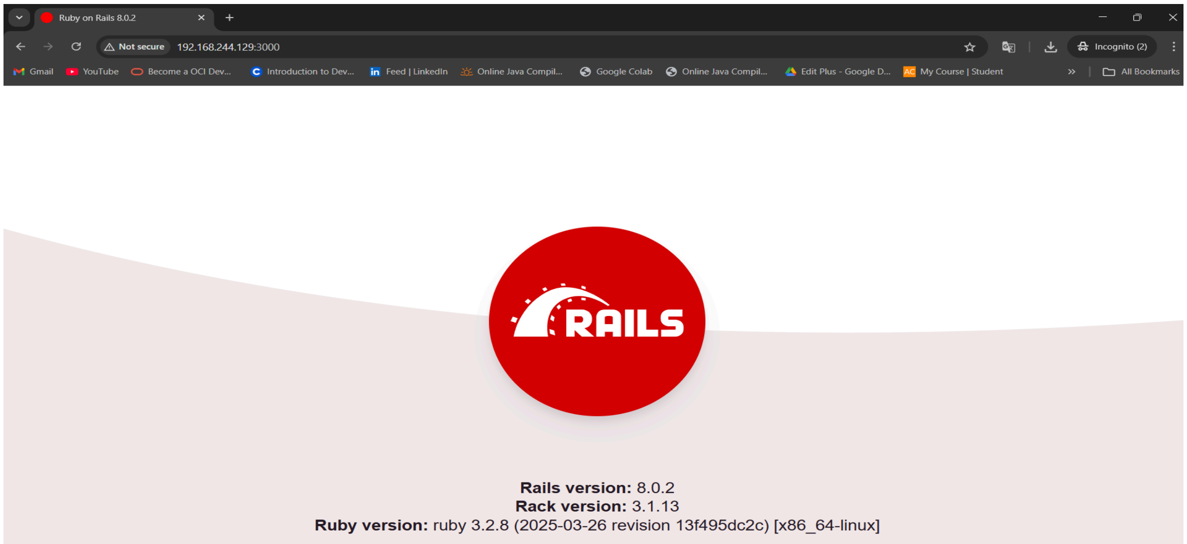
It is building and create a image push it on dockerhub and run container on docker in port 3000 which we mention in Dockerfile.



Here docker container and images which we build through CICD pipeline



It server on port 3000



Task 3: Deploy a Static Website Using Ansible

Objective: Serve a simple HTML page using Nginx, provisioned entirely via Ansible.

Requirements:

- **Install Nginx**
- **Deploy a custom `index.html` (Ansible template or copy module)**
- **Restart Nginx if config or content changes**
- **Handle rollback in case of failure (basic error handling)**

Completed:

Folder structure where stored index.html and playbook.yml for ansible.

```
vikas@admin:~/parity_Assignment/servepage_via_ansible$ tree
.
├── files
│   └── index.html
└── playbook.yml

2 directories, 2 files
vikas@admin:~/parity_Assignment/servepage_via_ansible$
```

When running ansible playbook it will install nginx configuration also check if it is install or not
Copy the index.html from files/ location to /var/www/html/index.html

Here is playbook.yml

```
vikas@admin:~/parity_Assignment/servepage_via_ansible$ cat playbook.yml
---
- name: Deploy a Static Website using Nginx
  hosts: 192.168.244.129
  become: yes  # Run tasks as root

  tasks:
    - name: Ensure Nginx is installed
      apt:
        name: nginx
        state: present
        update_cache: yes
      register: nginx_install
      retries: 3
      delay: 5
      until: nginx_install is succeeded

    - block:
        - name: Copy the custom index.html file
          copy:
            src: files/index.html  # Your local HTML file
            dest: /var/www/html/index.html
            owner: www-data
            group: www-data
            mode: '0644'
          notify:
            - Restart Nginx

        - name: Ensure Nginx service is started and enabled
          systemd:
            name: nginx
            state: started
            enabled: yes

      rescue:
        - name: Rollback Nginx configuration changes
          command: systemctl restart nginx
          ignore_errors: yes

        - name: Log failure message
          debug:
            msg: "An error occurred, and the Nginx configuration changes were rolled back."

      always:
        - name: Ensure Nginx is running
          systemd:
            name: nginx
            state: started
```

Here is execution output of Ansible playbook.

```
vikas@admin:~/parity_Assignment/servepage_via_ansible$ ansible-playbook playbook.yml -i 192.168.244.129, --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Deploy a Static Website using Nginx] ************************************************************************

TASK [Gathering Facts] ********************************************************************************************
ok: [192.168.244.129]

TASK [Ensure Nginx is installed] **********************************************************************************
ok: [192.168.244.129]

TASK [Copy the custom index.html file] ****************************************************************************
ok: [192.168.244.129]

TASK [Ensure Nginx service is started and enabled] ****************************************************************
ok: [192.168.244.129]

TASK [Ensure Nginx is running] ************************************************************************************
ok: [192.168.244.129]

PLAY RECAP ********************************************************************************************************
192.168.244.129            : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Web Ui